FINAL PROJECT REPORT

CARBON CONTROL

Carbon Footprint Reduction using Data Visualisation

Program	1 Million for 1 Billion (1M1B) Internship
Institution	Alliance University
Team Members	Mahir Mavani | Prem Panchal | Piyush Tandale
SDG Alignment	SDG 13 – Climate Action
Live Website	https://relaxed-rugelach-ed2c73.netlify.app/
Tableau Dashboard	https://public.tableau.com/app/profile/mahir.mavani/viz/CarbonEmission_17735177577970/

1. Executive Summary

Carbon Control is a student-led data visualization initiative developed as part of the 1M1B internship program at Alliance University. The project addresses a critical gap in public climate literacy — despite the abundance of carbon emissions data, most people lack the visual tools to interpret it meaningfully.

The team built an interactive Tableau dashboard and a live web-based interface that allows users to compare carbon emissions, renewable energy share, and fossil fuel dependency across five countries: India, Germany, Brazil, Canada, and Japan. The solution is publicly deployed and requires no technical expertise to use.

User testing confirmed that 100% of participants reported an improved understanding of carbon emissions after interacting with the dashboard — validating the project's core objective of making environmental data accessible, visual, and actionable.

2. Problem Statement

Communities, businesses, and individuals often lack clear visual insights into actions that reduce carbon emissions. Without accessible and understandable data, it becomes difficult to adopt sustainable practices.

Specifically, the following challenges were identified:

- Climate data exists in abundance but is typically locked in complex reports and raw datasets inaccessible to non-technical audiences.
- There is no simple, free, publicly available tool that lets an everyday user compare carbon emission trends across multiple countries.
- Awareness of low-carbon strategies — such as renewable energy adoption — remains low despite being critical to climate action.
- Data-driven tools such as dashboards can help visualize carbon emissions and energy patterns, enabling better understanding of low-carbon initiatives.

3. Project Objectives

The project was guided by the following core objectives:

- Analyze global energy consumption and carbon emissions data across multiple countries.
- Identify patterns related to renewable energy adoption and fossil fuel dependency over time.
- Develop an interactive dashboard to visualize key sustainability indicators in a user-friendly format.
- Promote awareness of low-carbon strategies using data visualization as the primary medium.
- Deploy a publicly accessible web interface so the tool reaches the widest possible audience.

4. Team & Roles

Team Member	Role	Responsibilities
Mahir Mavani	Data Visualization Lead	Tableau dashboard design, chart creation, publishing on Tableau Public
Piyush Tandale	Data Analyst	Dataset sourcing from Kaggle, data cleaning, filtering, and structuring for visualization
Prem Panchal	Web Developer	HTML/CSS web interface development, Netlify deployment, dashboard integration

5. Dataset Used

Dataset Name	Global Energy Consumption Dataset
Source	Kaggle
Countries	India, Germany, Brazil, Canada, Japan
Key Variable 1	Carbon Emissions (Million Tons)
Key Variable 2	Renewable Energy Share (%)
Key Variable 3	Fossil Fuel Dependency (%)
Key Variable 4	Total Energy Consumption
Key Variable 5	Country & Year

6. Tools & Technologies

Category	Tool / Technology	Purpose
Data Visualization	Tableau Public	Interactive dashboard development and public hosting
Data Preparation	Excel / CSV	Data storage, cleaning, and structuring
Data Source	Kaggle	Global Energy Consumption Dataset sourcing
Web Development	HTML & CSS	Web interface design and layout
Deployment	Netlify	Live hosting and public deployment of website

Category	Tool / Technology	Purpose
Presentation	Canva	Visual design of project presentation and report slides

7. Methodology & Work Breakdown Structure

The project followed a structured six-phase methodology from case study selection through to final deployment and sharing:

Phase	Activity	Description
1	Case Study Selection	Selected five countries — India, Germany, Brazil, Canada, and Japan — as case studies to analyze diverse energy patterns across different economic and geographic contexts.
2	Data Collection	Sourced the Global Energy Consumption dataset from Kaggle containing multi-year data on carbon emissions, renewable energy share, fossil fuel dependency, and total energy consumption.
3	Data Cleaning & Structuring	Filtered relevant countries and indicators, removed inconsistencies, and structured the dataset into a clean format ready for Tableau visualization.
4	Dashboard Development	Built interactive charts in Tableau Public showing trends in emissions, renewable energy share, and fossil fuel dependency. Published the dashboard for public access.
5	Scenario Modeling	Analyzed how increased renewable energy adoption could help reduce fossil fuel dependency and overall carbon emissions across the selected countries.
6	Prototype & Deployment	Built a web-based interface using HTML and CSS, integrated the Tableau dashboard, and deployed it live on Netlify for public access and user testing.

8. Prototype Development

The team developed a two-component prototype consisting of a data visualization layer and a web interface layer:

8.1 Tableau Dashboard

- Built an interactive, multi-chart dashboard using Tableau Public.
- Dashboard allows users to compare all five countries side by side.

- Three primary visualizations: carbon emissions over time, renewable energy share, and fossil fuel dependency.
- Published publicly at Tableau Public for open access.

8.2 Live Web Interface

- Designed and developed a web interface using HTML and CSS.
- Integrated the Tableau dashboard directly into the website for seamless user experience.
- Deployed on Netlify — fully functional and publicly accessible.
- Designed for intuitive navigation, requiring no technical background from the user.

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9. User Testing

The prototype was tested with multiple users — primarily students and peers — to evaluate usability, comprehension, and design effectiveness. Users were asked to perform three core tasks:

- Compare carbon emission data across countries.
- Analyze emission trends over time.
- Explore renewable energy share data.

10. User Feedback & Results

Feedback Parameter	Result
Primary Age Group	18–25 years
Usability Rating	4–5 out of 5 (Easy to navigate)
Design Rating	4–5 out of 5 (Clean and visual)
Improved Understanding	100% of users reported improved understanding of carbon emissions
Awareness Increase	Increased for ALL users
Information Clarity	Easy to understand for most users
Future Use Intent	Mostly Yes, few Maybe
Best Features Cited	Graphs, country comparisons, ease of use
Suggested Improvements	Real-time data, improved mobile UI, better interface design

11. Impact Created

11.1 Quantitative Impact

- 5 countries analyzed with real-world sustainability data
- 3 key indicators tracked — carbon emissions, renewable energy share, fossil fuel dependency
- 1 interactive Tableau dashboard built and publicly deployed
- 1 live website launched and hosted on Netlify
- 100% of users reported improved understanding of carbon emissions
- 4–5 out of 5 average rating on usability and design from all testers
- Multiple user testing sessions conducted with students and peers

11.2 Qualitative Impact

- Increased awareness of carbon emissions and low-carbon energy strategies among users.
- Made complex climate data visually accessible to non-technical audiences.
- Encouraged behavioural curiosity — users expressed intent to explore sustainability data further.
- Created a free, publicly available resource usable by educators, students, and researchers.
- Demonstrated how student-led technology solutions can meaningfully contribute to climate conversations.
- Sparked discussion around renewable energy adoption versus fossil fuel dependency among peer testers.

12. Learnings from the Internship

12.1 Green Skills

The team developed a hands-on understanding of what carbon emissions mean at a country level — how fossil fuel dependency drives emissions, and how renewable energy adoption shifts that trajectory. Working with real energy data made climate change feel measurable and urgent rather than abstract.

12.2 Data Skills

The team learned to source, clean, and structure a real-world dataset from Kaggle for practical use. Working with Tableau deepened understanding of how raw numbers become visual stories — through chart selection, data filtering, and dashboard design choices that directly affect how people interpret information.

12.3 Professional Skills

The internship reinforced the value of defined team roles, shared ownership, and iterative development. Gathering user feedback and refining the prototype based on real responses gave the team experience in the product development cycle. Most importantly, the team learned that clarity and communication are as critical as technical execution.

13. Next Steps & Future Scope

- Integrate real-time carbon emissions data via APIs to keep the dashboard current.
- Expand coverage to additional countries and regions for broader global comparison.
- Improve mobile responsiveness of the web interface based on user feedback.
- Enhance UI/UX with a more polished visual design and better navigation.
- Explore predictive modeling — forecast emission trends based on current energy transition rates.
- Seek partnerships with educational institutions to use Carbon Control as a classroom learning tool.

14. Conclusion

Carbon Control is not just a dashboard — it is a data-driven decision-support system that enables users to explore carbon emissions and sustainability trends in an intuitive, visual format. Through prototype development, real user testing, and iterative feedback analysis, the project demonstrates how technology can bridge the gap between complex environmental data and public understanding.

The project reinforces the role of digital solutions in driving awareness and encouraging low-carbon, sustainable practices. It proves that student-led initiatives, when grounded in real data and built with a user-first mindset, can make a meaningful contribution to global climate conversations.

Carbon Control — Visualize. Understand. Act.